

SLICES-RI Post-5G Blueprint hands-on

- **Duration:** 6 Hours (excluding lunch)
 - **Format:** Hands-on Tutorial (5G + docker + measurements)
-

Core Learning Outcome

By the end of the day, participants will be able to deploy a 5G network composed of open-source 5G core and RAN implementations. They will also be able to run traffic workloads on the network from multiple UEs, as well as collect and analyze control-plane and user-plane traffic.

Morning Session: Background on 5G and on reproducibility (3 Hours)

- 9:30 - 11:00 | Module 1: Foundations - 5G principles and architecture

Focus: 5G architecture, understanding the interactions of the different functions composing the core and the RAN

- 11:00 - 11:15 | Coffee Break
- 11:15 - 12:45 | Module 2: Reproducible experimental research with the SLICES-RI infrastructure.

Focus: Understand and prepare a SLICES-RI environment to run experiments.

- **15 min:** Introduction to SLICES-RI infrastructure.
- **30 min:** Hands-on: Setup a research environment.
- **45 min:** Hands-on: running a first (hello-world) experiment with data collection and publication.

Afternoon Session: 5G Experiments (3 Hours)

- 13:45 - 15:15 | Module 3: Deploying a 5G network and run an experiment on it.

Focus: Deploy your own 5G network with SLICES-RI.

- **30 min:** Hands-on: Configure and deploy a 5G core.
- **30 min:** Hands-on: Configure and deploy a 5G RAN.
- **30 min:** Hands-on: Analyse signaling traffic and compare it to theory.

- 15:15 - 15:30 | Coffee Break
- 13:45 - 15:15 | Module 4: Traffic workload on 5G and analysis.

Focus: Connect UEs to the 5G network and run experiments on it.

- **20 min:** Hands-on: Configure and deploy UEs.
- **1h10:** Hands-on: Run workload on the network, participants will decide their own scenario and analysis.

Required Competencies and Materials

Technical Prerequisites

- **Linux command line:** Intermediate-level knowledge of Linux command line. Participants should be comfortable with using the Linux command line.
- **Basic Network Concepts:** A rudimentary understanding of how network work in general will be beneficial, though core concepts will be covered.

Software and Setup

A preconfigured environment will be made available on the SLICES RI.

On their laptop, participants will require Python 3.10+ and SSH to prepare their experimental infrastructure and access it.